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|                  |                                           |
|------------------|-------------------------------------------|
| <b>Status</b>    | Finished                                  |
| <b>Started</b>   | Thursday, 30 October 2025, 1:31 PM        |
| <b>Completed</b> | Thursday, 30 October 2025, 1:44 PM        |
| <b>Duration</b>  | 13 mins 31 secs                           |
| <b>Marks</b>     | 1.25/3.00                                 |
| <b>Grade</b>     | <b>1.25</b> out of 3.00 ( <b>41.69%</b> ) |

Information

## Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

## Question 1

Incorrect

Mark 0.00 out of 1.00

Let  $A$  and  $X$  be the sets

$$A = \{a, b, c, d, e\}$$

$$X = \{!, *, +, -\}.$$

In this problem we consider words written in the alphabet which is the union of the two sets:  $A \cup X$ . This means that a word in this alphabet can contain letters from both  $A$  and  $X$ .

However, not all words are valid. The following rules apply:

1. A word must start with a letter from the set  $A$ .
2. A word must end with a letter from the set  $A$ .
3. A word can not contain two adjacent letters from the set  $X$ .

For example the word **a+baa!c** is valid, but the words **a+baa+!c**, **!aaa+c** and **acca!** are not.

The length of a word is the number of letters it consists of. For example, the length of **a+baa!c** is 7.

Let  $a_n$  equal the number of valid words of length  $n$ .

**a)** Find a recurrence relation satisfied by  $a_n$ :

$$4 \cdot a[n+1] - 3 \cdot a[n-2] = 2$$

Your last answer was interpreted as follows:

$$4 \cdot a_{n+1} - 3 \cdot a_{n-2} = 2$$

The variables found in your answer were:  $[a_{n-2}, a_{n+1}]$

**Syntax help:** To express the symbol " $a_n$ " you should enter **a[n]**. For example, if you think the correct answer is  $a_{n+1} - 2a_{n-3} = 1$ , you should answer **a[n+1] - 2\*a[n-3] = 1**.

**b)** Compute the value

$$a_1 = 4$$

Your last answer was interpreted as follows:

$$4$$

**c)** Compute the value

$$a_3 = 12$$

Your last answer was interpreted as follows:

$$12$$

**d)** Find the general solution to the recurrence relation satisfied by  $a_n$ .

$a_n =$  

**Syntax help:** You are free to use constants in your answer. For example, if you think the correct answer is  $2 \cdot n - A \cdot 3^n + B$ , you should answer **2\*n - A\*3^n + B**.

## Question 2

Partially correct

Mark 0.50 out of 1.00

In this problem you are tasked with computing with elements in the symmetric group of permutations,  $(S_5, \cdot)$ .

**Syntax guide:** The answer to each task is an element of the group  $S_5$  and should be given as a  $2 \times 5$  matrix. E.g. if you think the answer to a question is the permutation denoted by

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

then you should answer

|   |   |   |   |   |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| 2 | 3 | 1 | 5 | 4 |

a)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 3 & 4 & 5 & 2 \end{bmatrix}$$

b)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 1 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 2 & 1 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 5 & 1 & 2 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 3 & 5 & 1 & 2 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 2 & 5 & 4 & 3 \end{bmatrix}$$

d)

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{bmatrix}$$

Your last answer was interpreted as follows:

$$\begin{bmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 4 & 3 & 2 \end{bmatrix}$$

Question **3**

Partially correct

Mark 0.75 out of 1.00

In this problem you are tasked with computing products and inverses in the group  $(P_{49}, \bullet_{49})$  of integers and multiplication modulo 49.

$25 \bullet_{49} 30 =$

Your last answer was interpreted as follows:

15

$30 \bullet_{49} 43 =$

Your last answer was interpreted as follows:

16

$43 \bullet_{49} 29 =$

Your last answer was interpreted as follows:

22

$25^{-1} =$

Your last answer was interpreted as follows:

49

Question **4**

Correct

Mark 0.00 out of 0.00

## Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 254

Signing code:

Your last answer was interpreted as follows:

14852

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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